



Netramind

STANDARD OPERATING PROCEDURE	
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Title: GxP Data Integrity

1 PURPOSE

This SOP defines requirements to ensure that data generated, processed, and reported by Netramind are accurate, complete, attributable, and protected from unauthorized modification, in support of GxP-regulated clinical trial activities..

2 SCOPE

This procedure applies to all systems and processes used by Netramind to ingest, process, analyze, review, and report data used for GxP decision-making in clinical trials, including ophthalmic image data, derived outputs, and associated metadata.

3 REFERENCES

- European Medicines Agency (EMA): EudraLex Volume 4, Part 1: Basic Requirements for Medicinal Products; Annex 11: Computerized Systems
- European Medicines Agency (EMA): Guideline on computerized systems and electronic data in clinical trials
- Food and Drug Administration (FDA): 21 CFR Part 11; 21 CFR parts 210 and 211 and 820
- International Conference on Harmonization (ICH) Q7: Good Manufacturing Practice Guide for Active Pharmaceutical Ingredients; November 2000
- International Conference on Harmonization (ICH) E6 (R3) Good Clinical Practice

4 DEFINITIONS

- 4.1 Audit Trail— Metadata records that allow the appropriate evaluation of the course of events by capturing details on actions (manual or automated) performed relating to information and data collection and, where applicable, to activities in computerized systems. The audit trail should show activities, initial entry and changes to data fields or records, by whom, when and, where applicable, why. In computerized systems, the audit trail should be secure, computer-generated and time stamped. Audit trails must be securely retained and accessible for review; centralized storage is preferred where feasible.
- 4.2 Business Process— A step or set of steps that make up a process performed within the quality system (e.g., manufacturing process, CAPA, laboratory method execution).
- 4.3 Certified Copy— A copy (irrespective of the type of media used) of the original record that has been verified (i.e., by a dated signature or by generation through a validated process) to have the same information as the original, including relevant metadata, where applicable.
- 4.4 Computerized System Validation— A process of establishing and documenting that the specified requirements of a computerised system can be consistently fulfilled from design until



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decommissioning of the system or transition to a new system. The approach to validation should be based on a risk assessment that takes into consideration the intended use of the system and the potential of the system to affect trial participant protection and the reliability of trial results.

- 4.5 Critical Data— Data collected from a clinical study that is important from a subject safety and data integrity standpoint.
- 4.6 Data Acquisition Tool (DAT)—A paper or electronic tool designed to collect data and associated metadata from a data originator in a clinical trial according to the protocol and to report the data to the sponsor. For Netramind, the sponsor refers to the clinical trial sponsor. Netramind acts as a service provider. The data originator may be a human (e.g., the participant or trial staff), a machine (e.g., wearables and sensors), or a computer system from which the electronic transfer of data from one system to another has been undertaken (e.g., extraction of data from an electronic health record or laboratory system). Examples of DATs include but are not limited to CRFs, interactive response technologies, clinical outcome assessments, including patient-reported outcomes (PROs) and wearable devices, irrespective of the media used.
- 4.7 Data Integrity— Data integrity includes the degree to which data fulfil key criteria of being attributable, legible, contemporaneous, original, accurate, complete, secure and reliable such that data are fit for purpose.
- 4.8 GxP Records— Records associated with the development, manufacture, packaging, testing, marketing, and distribution of the active substance (i.e., API), pharmaceutical product (i.e., drug product), or medical device that are used for clinical and commercial purposes. This also includes registration and regulatory records for both clinical and commercial products.
- 4.9 Metadata— The contextual information required to understand a given data element. Metadata is structured information that describes, explains or otherwise makes it easier to retrieve, use or manage data. For the purpose of this guideline, relevant metadata are those needed to allow the appropriate evaluation of the trial conduct.
- 4.10 Original record/data— Data as the file or format in which it was originally generated, preserving the integrity (accuracy, completeness, content and meaning) of the record, e.g., original paper record of manual observation, or electronic raw data file from a computerized system. (Source: Data Integrity Definitions and Expectations (Revision 1), Jan, 2015, MHRA)
- 4.11 Raw Data— Factual recorded information obtained as a result of original observations or activities, which has not been copied or transcribed. This term is synonymous with original data.
- 4.12 Source Records— Original documents or data (which includes relevant metadata) or certified copies of the original documents or data, irrespective of the media used. This may include trial



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participants' medical/health records/notes/charts; data provided/entered by trial participants (e.g., electronic patient-reported outcomes (ePROs)); healthcare professionals' records from pharmacies, laboratories and other facilities involved in the clinical trial; and data from automated instruments, such as wearables and sensors.

4.13 System Administrator— A person who, based on training, experience, and education, is granted higher level of system access to properly maintain the system. This administrator may include a System Administrator, Application Administrator, or a Business Administrator.

4.14 System / Process Owner— The person ultimately responsible for the business process or processes being managed.

4.15 System of Record— A type of software-based application, such as an electronic quality management system, that maintain or manage a formalized system that defines and documents processes, procedures, and responsibilities for achieving quality policies, practices, and objectives.

5 RESPONSIBILITIES

5.1 System or Process Owner

- 5.1.1 Ensure that systems and procedures are in place to ensure data integrity from creation and generation of the data up until destruction.
- 5.1.2 Ensure service suppliers are on the approved supplier list and approved suppliers are aware of the requirements in this procedure.
- 5.1.3 Ensure data flows, transformations, and outputs are documented and traceable.
- 5.1.4 Determine audit trail needs.

5.2 Quality

- 5.2.1 Define the audit trail transactions to comply with ICH standards and audit trail review requirements.
- 5.2.2 Determine when to review the audit trail data, which includes the relevant metadata.
- 5.2.3 Define audit trail review expectations based on data criticality.

5.3 End User

- 5.3.1 Follow systems and procedures in place to ensure data integrity.
- 5.3.2 Notify the System / Process Owner if data integrity issues are suspected or observed.

6 PROCEDURE

6.1 General Requirements

- 6.1.1 The End Users' management must ensure that the personnel have the appropriate training to handle, document, manage, manipulate, and/or review data.
- 6.1.2 Record-keeping practices for data that meet predicate rule under any current GxP regulations must be in place to prevent data from being lost or obscured.



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- 6.1.3 Any data created as part of a GxP record must be evaluated by Quality prior to use in GxP decision-making or external reporting.
- 6.1.4 The System / Process owner must ensure that suppliers are on the approved supplier list and that the supplier assessments, including assessments for third-party service providers (e.g., laboratory contractors that provide results used in GxP decision making), were performed to ensure that the suppliers were reviewed for intended use.
- 6.1.5 The segregation of duties must be maintained.
 - 6.1.5.1 The data reviewers/approvers cannot be the same people that generate or manipulate the data.
 - 6.1.5.2 For electronic systems, Administrators should be independent of the data's End Users.
- 6.1.6 Computerized systems used to create, maintain, store versions, approve, transmit and distribute GxP documents and records must be designed and validated for use, according to IT procedures.
- 6.1.7 Critical data includes, but is not limited to: image-derived measurements, grading outputs, adjudication decisions, and datasets provided to sponsors.
- 6.2 Hybrid Systems
 - 6.2.1 When data printouts (i.e., paper format) from an electronic system are used as the system of record, the printout must be accurate, reliable complete, and preserve the content and meaning of the data.
 - 6.2.2 The data review conducted by System / Process owner and Quality on the printout must include a determination that the data accurately represents the source data.
 - 6.2.3 The review must be documented on the record.

NOTE: If the printout/report functionality has been validated, then this review may not be required.
- 6.3 Electronic data integrity requirements
 - 6.3.1 The System / Process owner's management must ensure that the systems used to generate, manipulate, or report on data used to make GxP decisions are validated for their intended use.

NOTE: If ad hoc queries are used, the query must be included in reporting that query so that another user can recreate the same data.
 - 6.3.2 Security controls must be in place for computerized systems that generate or maintain data used for GxP functions.
 - 6.3.2.1 Procedures for controlling access rights to the computerized systems must be documented.



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- 6.3.3 Systems that allow users to edit the originally entered data and used to generate, manipulate, or report on data must be configured to provide the audit trail most appropriate for the system and business process.
NOTE: Audit trail data may not always be found in the audit trail, and instead may be found in applicable metadata or event logs.
- 6.3.4 Data should be maintained throughout the record's retention period with all associated metadata required to reconstruct any series of data modification (as applicable) for a record.
- 6.3.5 The relationships between data and their metadata should be preserved in a secure and traceable manner.
- 6.3.5.1 The audit trail configuration must be risk-based. For example, data that is used to make direct GxP decisions must have an audit trail that captures:
- Name of a person,
 - Date/time of the change,
 - Data that changed,
 - Reason for change (if such information is not obvious from the record)
- 6.3.5.2 End Users and data reviewers/approvers must be able to readily and accurately determine when source data was changed, who changed it, and what the original data was.
- 6.3.5.3 The reason for the change must be linked to the data when it is not otherwise obvious.
NOTE: Changes to the approved source data must be performed under an approved change control process.
- 6.3.5.4 When GxP critical data is entered manually, the system should contain error detection as applicable for:
- Invalid field entries
 - Values out of range
 - Blank field if entry is required
 - Invalid data type checks on step sequencing
- 6.3.6 When GxP critical data is entered manually, the accuracy and completeness of the data must be verified. Manual data entry controls should be applied based on risk and frequency of manual input.
- 6.3.6.1 Depending on the data's criticality and the probability of an error occurring, manual data entry may be managed by one of the following mechanisms (listed from the most to the least secure):



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- Double-blind data entry in which two personnel enter the same data without seeing what the other has entered. The system then checks for consistency between the two data entry sets.
 - Data entry that is verified by a second person authorized to perform the verification.
 - Documented check and review using a printout of the data before it can be used electronically.
 - Electronic approval of the data entry and/or the complete record using electronic signatures.
- 6.3.7 For data that are considered non-critical, a risk-based approach should be undertaken to ensure data integrity of the data.
- 6.3.8 Processing or manipulation of data.
- 6.3.8.1 For algorithm-generated outputs, the version of the algorithm used must be recorded and traceable.
- 6.3.8.2 Changes to data can only be performed by authorized personnel and must be traceable as defined by a Standard Operating Procedure (SOP).
- 6.3.8.3 The System / Process owner must document, and control data sources used for calculation or downstream manipulation.
- 6.3.8.4 The System / Process owner must define when and how data may be further processed. (e.g., once the raw data is approved)
- 6.3.8.5 The following must be documented when reporting data in a format different from how the raw data was recorded:
- The source, or the origination of the raw data
 - How the data was manipulated, sorted, charted, or reported
 - Who performed the manipulation
- NOTE: If the format change is a unit conversion, this can be achieved by ensuring both the original data and the newly formatted data contain the unit of measure. Therefore, later review can confirm the data conversion
- 6.3.8.6 If the data requires a review, the review must be performed by independent personnel that can confirm the accuracy of the source data.
- 6.3.9 Interfaces
- 6.3.9.1 Data transfers between systems must be verifiable, with reconciliation where applicable.
- 6.3.9.2 For any system interfaces, there must be documentation of what data is being pulled from / pushed into other systems.
- 6.3.9.3 There must be documented confirmation of its accuracy, including significant figures, and rounding rules.



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NOTE: The degree and frequency of the input/output verification should be based on the complexity of the interface of the I/O devices with the computer system.

- 6.3.9.4 There must be controls and documentation of any human interface data manipulation.
- 6.3.9.5 Data must be controlled/validated or otherwise approved/verified before it is sent to another system or source.
- 6.3.9.6 The system of record for all critical data must be clearly defined and documented.
- 6.3.10 Data archiving and retrieval
 - 6.3.10.1 Archived files must be complete and accurate to preserve the content and meaning of the original file. For example, may include raw data files, result data files, as well as metadata and audit trail files.
- 6.3.11 Data backup and recovery
 - 6.3.11.1 Duplicate or data backup systems must be designed to ensure that the backed-up data is exact and complete, and that it is secure from modification, destruction, or loss. (CFR 211.68)
- 6.3.12 Audit trail review
 - 6.3.12.1 System / Process owner and Quality define which transactions are important to review.
 - 6.3.12.2 System / Process owner and Quality determine where the data can be found. (e.g., audit trail file, metadata file, event logs)
 - 6.3.12.3 Audit trail review frequency should be risk-based and aligned to data criticality.
 - 6.3.12.4 System / Process owner and Quality define when to review the data (e.g., routinely, per record, or periodically based on a sample plan per a specified time period) and who performs the review.

7 ATTACHMENTS

- N/A



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8 REVISION HISTORY

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01	01-MAY-2026	New Document
[X.0]	[DD-MMM-YYYY]	[Insert a summary of the changes]



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SIGNATURE PAGE

[ADD ADDITIONAL ROWS AS NEEDED]

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